

TABLE 3 Median Own and Cross-Price Elasticities

#	Brand	K Corn Flakes	K Raisin Bran	K Frosted Flakes	K Rice Krispies
1	K Corn Flakes	-3.696	.023	.500	.010
2	K Raisin Bran	.023	-2.061	.088	.051
3	K Frosted Flakes	.361	.059	-3.546	.028
4	K Rice Krispies	.010	.048	.040	-1.320
5	K Frosted Mini Wheats	.000	.053	.003	.057
6	K Froot Loops	.000	.010	.008	.038
7	K Special K	.155	.072	.248	.039
8	K NutriGrain	.270	.094	.313	.023
9	K Crispix	.003	.038	.020	.079
10	K Cracklin Oat Bran	.000	.023	.001	.046
11	GM Cheerios	.007	.080	.035	.069
12	GM Honey Nut Cheerios	.001	.017	.017	.043
13	GM Wheaties	.503	.113	.445	.029
14	GM Total	.140	.064	.238	.042
15	GM Lucky Charms	.000	.012	.010	.041
16	GM Trix	.000	.010	.009	.043
17	GM Raisin Nut	.007	.137	.043	.059
18	P Raisin Bran	.014	.232	.063	.050
19	P Grape Nuts	.001	.048	.006	.050
20	Q 100% Natural	.000	.023	.002	.048
21	Q Life	.003	.038	.052	.048
22	Q CapNCrunch	.001	.013	.015	.038
23	R Chex	.005	.037	.028	.081
24	N Shredded Wheat	.002	.081	.018	.049
25	Outside good	.158	.036	.107	.036

TABLE 3 Extended

GM Cheerios	GM Lucky Charms	P Raisin Bran	P Grape Nuts	Q Life	R Chex	Shredded Wheat
.011	.000	.007	.000	.000	.002	.001
.131	.008	.110	.030	.011	.012	.032
.040	.004	.021	.003	.008	.006	.005
.106	.025	.023	.030	.033	.024	.018
.102	.034	.032	.054	.044	.015	.023
.043	.111	.006	.024	.163	.010	.008
.054	.002	.025	.019	.004	.009	.028
.046	.001	.034	.011	.002	.004	.022
.103	.027	.019	.032	.035	.024	.018
.103	.040	.014	.039	.058	.012	.011
-1.709	.020	.041	.037	.028	.021	.024
.055	.099	.009	.026	.142	.012	.010
.054	.001	.041	.003	.002	.005	.007
.059	.003	.022	.020	.005	.010	.027
.049	-1.945	.007	.026	.149	.011	.009
.052	.102	.006	.024	.141	.012	.009
.160	.012	.065	.029	.019	.016	.026
.134	.009	-2.030	.036	.012	.011	.034
.089	.025	.026	-2.096	.032	.013	.070
.103	0.42	.013	.036	.063	.013	.011
.080	.072	.019	.028	.103	.014	.015
.048	.105	.007	.023	-1.559	.010	.008
.106	.024	.017	.027	.031	-1.749	.017
.099	.015	.043	.115	.020	.014	-2.268
.048	.017	.016	.017	.030	.009	.010

Knittel and Metaxoglou

(Review of Economics and Statistics, forthcoming)

- A cautionary tale of estimating random coefficients logit discrete choice models without carefully addressing computational complications
- Vary computation:
 - Specific algorithm (e.g gradient-based, direct search with function evaluations)
 - Starting values
 - Tolerances
- This is well known in the nonlinear optimization literature, but not internalized by practitioners

Table 1: Papers using BLP-type demand models with aggregate data in selected journals

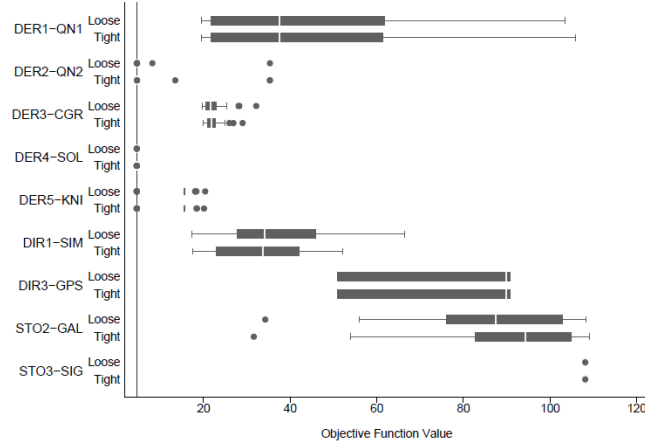
Journal	Paper
American Economic Review	Berry, Levinsohn and Pakes (1999)
Econometrica	Berry, Levinsohn and Pakes (1995)
	Goeree (2008)
	Nevo (2001)
Journal of Political Economy	Berry, Levinsohn and Pakes (2004)
	Petrin (2002)
Review of Economics and Statistics	Nevo (2003)
	Rekkas (2007)
Review of Economic Studies	Berry, Levinsohn and Pakes (2004)
	Nakamura and Zerom (2010)
	Villas-Boas (2007)
Rand Journal of Economics	Akerberg and Rysman (2005)
	Armantier and Richard (2008)
	Berry (1994)
	Bonnet and Dubois (2010)
	Chu (2010)
	Copeland (2011)
	Davis (2006)
	Iizuka (2007)
	Nevo (2007)
	Song (2007)
	Villas-Boas (2009)

Note: The list of journals is limited to the leading general interest and industrial organization journals. In addition, only papers that contain the main ingredients of the BLP approach for the estimation of random-coefficient Logit models using aggregated data as discussed in Section 1 are included.

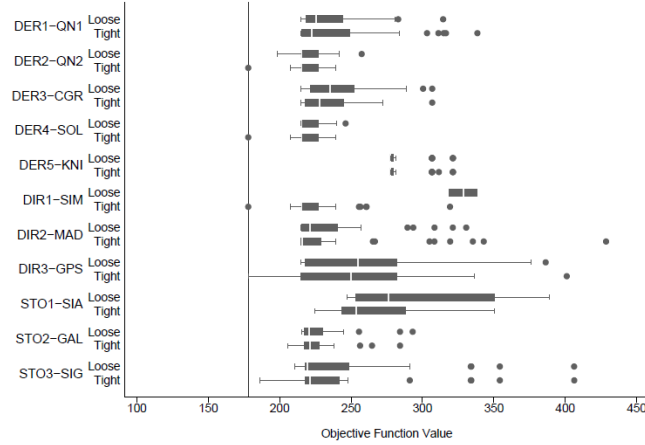
Big Picture

- Sections 2-3 are nice succinct summaries of Nevo's "Practitioner's Guide..." and Nevo's merger papers
- Applies standard estimation methods (Nested Fixed Point) to 2 classic datasets
 - Nevo (2000) – cereals
 - Berry et al. (1995) – automobiles
- Other method available: Mathematical Program with Equilibrium Constraints (MPEC)
 - See e.g. Su and Judd (2011)

Figure 1: Box-and-Whisker plots of the objective function value by optimization algorithm and NFP tolerance for cereals and automobiles



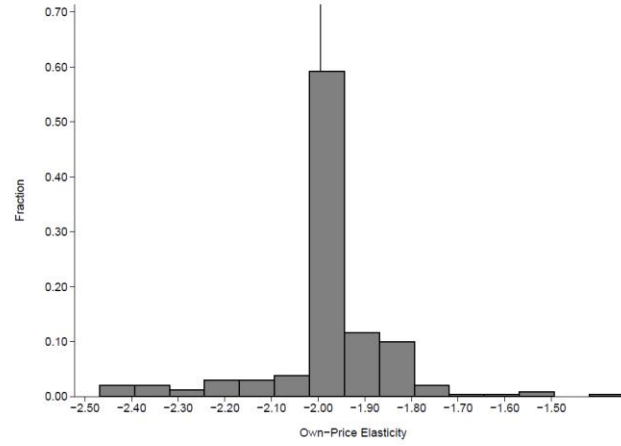
(a) Cereals



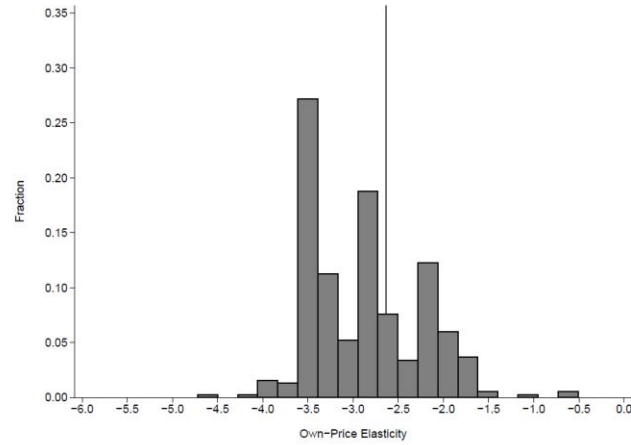
(b) Automobiles

Note: The naming convention of the optimization algorithms on the vertical axis of the Box-and-Whisker plots (BaW) in this figure follows Table 2. Loose and Tight refer to the alternative nested fixed-point (NFP) tolerances described in Section 4. The vertical solid line indicates the minimum objective function value for the combinations of optimization algorithms, NFP tolerances, and starting values that converged. We implemented DER5-KNI using a lower bound of zero on the parameter vector over which the algorithm performed the nonlinear search.

Figure 2: Own-price elasticity histogram of the top product with tight NFP tolerance for cereals and automobiles



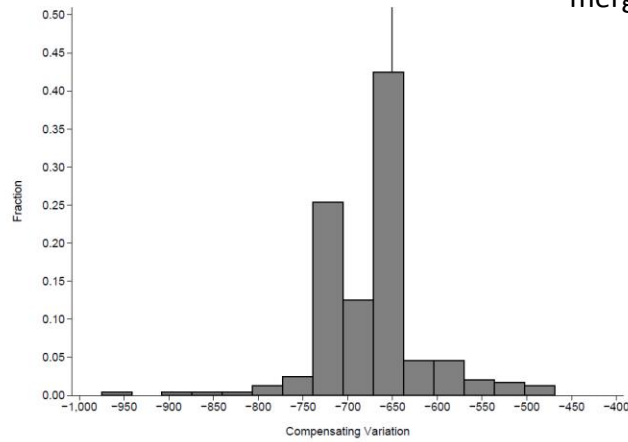
(a) Cereals



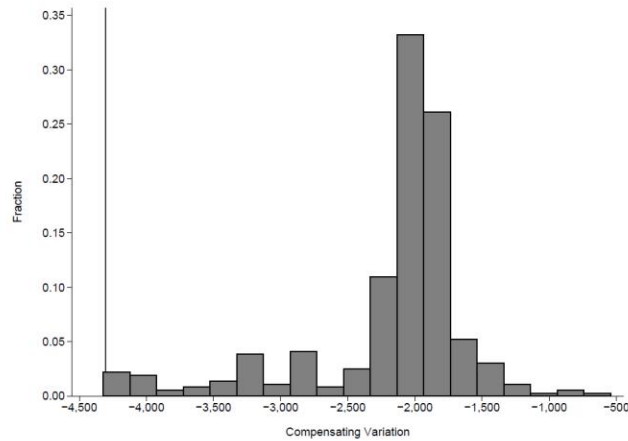
(b) Automobiles

Note: The top product is the product with the largest observed unit market share. For cereals, the top product is brand 5 in market 53 with market share of 45%. For automobiles, the top product is BKRIVE72 in 1972 with market share of 1%. The histograms are based on 240 (383) observations in the case of cereals (automobiles). Observations exceeding 0 are excluded in the case of automobiles. The number of observations reflects the sets of starting values that allowed the optimization algorithms to converge under tight NFP tolerance, the removal of extraneous observations discussed in Section 7.1, and any thresholds discussed here. The vertical lines indicate the value of the own-price elasticity when the objective function value is 4.56 (178.06) for cereal (automobiles).

Figure 7: Average market compensating variation for hypothetical ~~variation~~ merger



(a) Cereals



(b) Automobiles

Note: The histogram is based on 240 (364) observations in the case of cereals (automobiles). In the case of automobiles, values below -4,321m (5th percentile) or above 0 are excluded. Each observation corresponds to a weighted average compensating variation across 94 (19) markets using pre-merger quantities of the inside goods as weights. The automobile market for 1971 is excluded due to extraneous observations. The number of observations reflects the sets of starting values that allowed the optimization algorithms to converge under tight NFP tolerance, the removal of extraneous observations discussed in Section 7.1, and any thresholds discussed here. The details of the compensating variation calculation are available in Section 7.3. The vertical lines indicate the value of the average market compensating variation when the objective function values is 4.56 (178.06) for cereal (automobiles).